

SQL Command Center — Documentación Técnica

Módulo: Index Advisor | **Objeto DB:** Vista SQL v_index_advisor

1. Definición Con Restricción (Producción)

Filtra tablas con más del 40% de escaneos secuenciales y exige coincidencia en pg_stat_statements.

```
CREATE OR REPLACE VIEW v_index_advisor AS
WITH tablas_criticas AS (
    SELECT
        schemaname || '.' || relname AS tabla,
        relid, seq_scan, seq_tup_read, idx_scan,
        pg_size_pretty(pg_relation_size(relid)) AS tamaño_tabla,
        ROUND((seq_scan::numeric / NULLIF(seq_scan + idx_scan, 0)) * 100, 2)::float4 AS pct_seq_scan
    FROM pg_stat_user_tables
    WHERE (seq_scan + idx_scan) > 50
        AND (seq_scan::numeric / NULLIF(seq_scan + idx_scan, 0)) > 0.40
),
consultas_asociadas AS (
    SELECT
        s.queryid, s.query::text AS consulta_sql, s.calls AS ejecuciones,
        ROUND(s.mean_exec_time::numeric, 2)::float4 AS tiempo_medio_ms,
        t.tabla, t.seq_scan, t.seq_tup_read, t.idx_scan, t.tamaño_tabla, t.pct_seq_scan,
        ROW_NUMBER() OVER (PARTITION BY t.tabla ORDER BY s.mean_exec_time DESC) AS rn
    FROM tablas_criticas t
    JOIN pg_stat_statements s ON s.query ILIKE '%' || split_part(t.tabla, '.', 2) || '%'
    WHERE s.query NOT ILIKE '%pg_stat%'
)
SELECT
    tabla, seq_scan, seq_tup_read, idx_scan, tamaño_tabla, pct_seq_scan,
    queryid, consulta_sql, ejecuciones, tiempo_medio_ms,
    'CREATE INDEX CONCURRENTLY idx_' || replace(split_part(tabla, '.', 2), '-', '_') ||
FROM consultas_asociadas WHERE rn = 1 ORDER BY seq_tup_read DESC;
```

2. Definición Sin Restricción (Testing & Debug)

Captura cualquier tabla con seq_scan > 0 usando LEFT JOIN y COALESCE.

```
DROP VIEW IF EXISTS v_index_advisor;

CREATE VIEW v_index_advisor AS
WITH tablas_criticas AS (
    SELECT
        schemaname || '.' || relname AS tabla,
        relname AS nombre_tabla,
        seq_scan, seq_tup_read, idx_scan,
        pg_size_pretty(pg_relation_size(relid)) AS tamaño_tabla,
        ROUND((seq_scan::numeric / NULLIF(seq_scan + idx_scan, 0)) * 100, 2)::float4 AS pct_seq_scan
    FROM pg_stat_user_tables
    WHERE seq_scan > 0
),
consultas_asociadas AS (
    SELECT
        t.tabla, t.seq_scan, t.seq_tup_read, t.idx_scan, t.tamaño_tabla, t.pct_seq_scan,
```

```
s.queryid, s.query::text AS consulta_sql, s.calls AS ejecuciones,
ROUND(s.mean_exec_time::numeric, 2)::float4 AS tiempo_medio_ms,
ROW_NUMBER() OVER (PARTITION BY t.tabla ORDER BY s.mean_exec_time DESC) AS rn
FROM tablas_criticas t
LEFT JOIN pg_stat_statements s ON s.query ILIKE '%' || t.nombre_tabla || '%' AND s.
)
SELECT
tabla, seq_scan, seq_tup_read, idx_scan, tamaño_tabla, pct_seq_scan,
COALESCE(queryid, 0) AS queryid,
COALESCE(consulta_sql, 'Sin consulta registrada en pg_stat_statements') AS consulta
COALESCE(ejecuciones, 0) AS ejecuciones,
COALESCE(tiempo_medio_ms, 0.0) AS tiempo_medio_ms,
'CREATE INDEX CONCURRENTLY idx_' || replace(split_part(tabla, '.', 2), '-', '_') ||
FROM consultas_asociadas WHERE rn = 1 OR rn IS NULL ORDER BY seq_tup_read DESC;
```

3. Cuadro Comparativo

Criterio	Versión Estricta	Versión Permisiva
Caso de Uso	Producción / Auditoría crítica	Desarrollo / Script de tests
Filtro Seq Scan	> 40% del total de accesos	> 0 escaneos acumulados
Relación pg_stat	INNER JOIN (Excluye sin match)	LEFT JOIN (Inclusivo)
◀ Manejo de Nulos	Descarta filas incompletas	Defaults mediante COALESCE ▶